Haonan Ke

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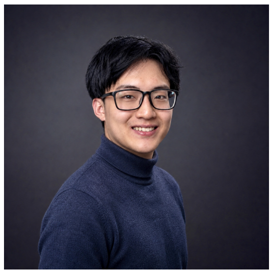

I am a master's student at the Technical University of Munich (TUM) with a dual focus in Electrical and Computer Engineering and Human Factors Engineering. My work sits at the intersection of edge AI, robotics, embedded systems, and hardware-aware computing, with a focus on real-time control and hardware/software co-design. At the same time, I am interested in bringing these technologies closer to people through human factors engineering, building intelligent products that are powerful, intuitive, and meaningful in real-world interaction.

Education

Technical University of Munich (TUM) — Munich, Germany
M.Sc. Electrical and Computer Engineering — *Oct. 2024 – Present*
Focus on embedded systems, autonomous systems, AI hardware.

Technical University of Munich (TUM) — Munich, Germany
M.Sc. Human Factors Engineering — *Oct. 2025 – Present*
Parallel master's program focusing on human-centered AI systems, HCI and HMI.

University of Virginia (UVA) — Charlottesville, USA
Exchange Semester, Electrical and Computer Engineering — *Aug. 2025 – Dec. 2025*
Recipient of Mühlfenzl Foundation scholarship.

Technical University of Munich (TUM) — Munich, Germany
B.Sc. Electrical and Computer Engineering — *Oct. 2020 – Apr. 2024*

Gymnasium Hohenschwangau — Schwangau, Germany
German Abitur — *Sep. 2017 – Jul. 2020*

Nanjing Foreign Language School (Xianlin Campus) — Nanjing, China
Secondary Education — *Sep. 2013 – Jun. 2017*

Professional Experience

Siemens AG — Munich, Germany
Working Student – Industrial Metaverse Lab — *Jun. 2026 – Present*
Contributing to industrial metaverse and XR prototypes, including interactive simulation, human-centered design, digital twin workflows, and robotics-related industrial applications.

Technical University of Munich — Munich, Germany
Research Intern – Chair of AI Processor Design — *Apr. 2025 – Oct. 2025*
Built a microcontroller-based memory characterization platform and automated SPI memory chip testing, covering hardware setup, signal validation, and operating-condition experiments.

IAV GmbH — Munich, Germany
Working Student – Vehicle Infotainment — *Jan. 2024 – Dec. 2024*
Developed Python and ECU-Test automation workflows for BMW Intelligent Personal Assistant validation, including voice-command testing, result analysis, and hardware-level test rack debugging.

AUDI AG — Ingolstadt, Germany
Intern – Vehicle Electronics — *Apr. 2023 – Sep. 2023*
Supported LIN/CAN-based ECU testing for vehicle electronics, including CAPL scripting, signal validation, protocol checks, diagnostic analysis, and test bench documentation.

Languages

Chinese, German, English